Kummer signed-pair uniqueness and transition obligations

2026-11-01. Source: preserved graph-integration candidate, not deployed-runtime certification.

Checked component. Three Lean declarations and three controls pass, with six separate axiom readbacks. Assuming the projection fibers are exactly $\{D, -D\}$ and $2G \neq 0$, matching both $\pi(D)$ and $\pi(G + D)$ uniquely identifies the signed divisor. A zero-anchor control proves the sign ambiguity when that condition is dropped. This matches the source candidate filter; the coordinate fiber property remains a geometric obligation.

Original-G invariant. For $D = aG + bQ$, inputs must represent $(\pi(D), \pi(D + G))$. Choice $c \in \{0, 1\}$ requires successor $(\pi(2D + cG), \pi(2D + (c + 1)G))$. The scalar recurrence alone does not establish these equations.

Transition branches. Ordinary addition, equal-input doubling, repaired addition, failed-pair signed fallback, and optional transition validation require separate obligations. The source calls xADD on equal inputs with difference $\pi(0)$; the literature's abstract xADD interface assumes unequal points. That specialization needs a geometric argument. Each repair action must implement rational two-torsion translation, square to projective identity and preserve nonzero vectors. Cross-product equality alone accepts zero vectors and cannot establish those premises.

Exact formula obligation. Let $W = H((H(X)_i H(Y)_i / t_i)_i)$ and $F_i = W_i^2 \prod_{j \neq i} Z_j$, matching source xadd. Prove $F_i \pi(P + Q)_j = F_j \pi(P + Q)_i$ for all indices and $F \neq 0$ on each claimed domain. Repeat the domain analysis for doubling and repaired calls.

Signed recovery branches. Generic Recover has special-anchor, $D = 0$, $\Delta = 0$ and $h_1 = 0$ exclusions. For its returned $U = X^2 - q_1 X + q_2$, $V = b_1 X + b_0$, prove $U \mid f - V^2$ and both projection matches. Exceptional enumerators require candidate completeness. The conditional uniqueness theorem handles the final matching-candidate step but not existence or formula soundness.

Next source-specific construction. Bind project_any and general_homogeneous as inverse projective maps on the chosen theta-model domain, then prove their quotient-fiber property for valid Mumford divisors. This supplies the missing geometric premise of the checked uniqueness result. Equal-input xADD specialization remains an immediate separate closure obligation.

Evidence limits. Existing cached helper binaries retain their trust boundary. Initial proof attempt and a wrong-working-directory invocation failed and are retained. Corrected proof, controls and readback pass. Independent semantic review is pending. No target enumeration, scientific execution or cost comparison occurred.

Primary reference: Chung, Costello and Smith, Fast, uniform, and compact scalar multiplication, sections 2 and 6 (Microsoft recovery.pdf). Detailed finite branch equations and source locations: REPORT.md. Editable graph, structured status, bindings and native receipts accompany this PDF.